

NORMALIZATION REPORT
BANK MANAGEMENT SYSTEM PROJECT

1. INTRODUCTION:

This document explains the normalization process used in the Bank Management System database project. The database has been designed to reduce redundancy, improve consistency, and maintain data integrity using normalization techniques up to BCNF.

The project includes multiple related tables such as Customers, Accounts, Transactions, Loans, and AuditLog. Proper primary key and foreign key relationships have been implemented to maintain referential integrity and efficient database management.

2. TABLES USED

The following tables are used in the project:

1. Customers
2. Accounts
3. Transactions
4. Loans
5. AuditLog

3. FIRST NORMAL FORM (1NF)

The database satisfies First Normal Form (1NF) because:

- All attributes contain atomic values.
- No repeating groups are present.
- Each column stores only one value.
- Each row is uniquely identified using primary keys.
- Duplicate rows are avoided.

Example:

Customer details such as FullName, Email, PhoneNumber, and Address are stored separately in single-value columns.

4. SECOND NORMAL FORM (2NF)

The database satisfies Second Normal Form (2NF) because:

- The database is already in 1NF.
- Partial dependency has been removed.
- Separate tables have been created for Customers, Accounts, Transactions, Loans, and AuditLog.
- Non-key attributes depend completely on the primary key.

Example:

Account details are stored separately from customer details, and transactions are stored separately from accounts to avoid unnecessary duplication.

5. THIRD NORMAL FORM (3NF)

The database satisfies Third Normal Form (3NF) because:

- The database is already in 2NF.
- Transitive dependency has been removed.
- Non-key attributes depend only on the primary key.
- Relationships between tables are maintained using foreign keys.

Example:

Customer information is stored only in the Customers table and referenced in other tables using CustomerID instead of storing duplicate customer details repeatedly.

6. BOYCE-CODD NORMAL FORM (BCNF)

The database satisfies Boyce-Codd Normal Form (BCNF) because every determinant in the database tables is a candidate key.

Proper primary key and foreign key relationships have been maintained throughout the database structure to ensure high-level normalization and data integrity.

7. RELATIONSHIPS USED IN THE PROJECT

The following relationships are used in the database:

- One Customer can have many Accounts.
- One Account can have many Transactions.
- One Customer can have many Loans.
- AuditLog table stores activity logs generated automatically using triggers.

8. ADVANTAGES OF NORMALIZATION

Advantages of normalization in this project include:

- Reduces data redundancy
- Improves data consistency
- Prevents insertion anomalies
- Prevents update anomalies
- Prevents deletion anomalies
- Improves database maintenance
- Enhances data integrity
- Makes database scalable and efficient

9. CONCLUSION

The Bank Management System database has been successfully normalized up to BCNF. The normalized structure improves efficiency, scalability, consistency, and integrity of the database system.

The project successfully implements core database concepts including:

- DDL Commands
- DML Commands
- Constraints
- Joins
- Subqueries
- Views
- Indexes
- Functions
- Triggers
- TCL Commands
- DCL Command

This database design provides a secure, scalable, and efficient banking management solution using Microsoft SQL Server.